Windows Domain Administration, Identity & Auditing

Seven support cases from my lab.local homelab

This is my first full Active Directory support project. I worked through seven small cases in my own VMs and kept the screenshots from each test. I tried to work the same way I would on a real ticket: check the current state, make one change, test it, and write down what actually happened. A few things did not behave the way I expected, so those are included too.

My lab

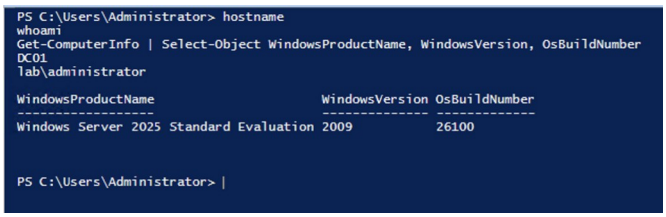
System / account	Role in the lab	Recorded state
DC01	Domain controller and DNS	Windows Server 2025 Standard Evaluation, 24H2, build 26100.32230. Static private lab IP.
WIN11-IT-01	Domain workstation	Windows 11 Pro 25H2, build 26200.9457. Computer object in the Workstations OU.
LAB\ajaradi	Normal user	Used for ordinary client work.
LAB\adm-ajaradi	Separate admin account	Used for UAC prompts and elevated admin tools.
LAB\jonas	Helpdesk test account	Used to test the Finance-only password-reset delegation.
WIN11-IT-01\LocalAdmin	Local fallback account	Checked before the trust test so I still had a local way into the PC.

Cases in this project

ID	Test	Status
P1-T1	Health, DNS and domain trust	**PASS**
P1-T2	Standard vs administrative identity	**PASS**
P1-T3	Logon auditing with Group Policy	**PASS**
P1-T4	Scoped helpdesk password reset	**PASS**
P1-T5	GPMC support incident	**PASS**
Break/Fix 1	Broken secure channel and repair	**PASS**
Break/Fix 2	Protected AD object deletion	**PASS**

Task: Before I changed anything, I wanted to confirm the server/client versions, where the PC sits in AD, and whether the workstation still trusts the domain.

I started by checking the machines before changing anything. DC01 is the domain controller and WIN11-IT-01 is the client I use for the tests. I used winver as the final check for the Windows product and build numbers so the versions are clear in the evidence.



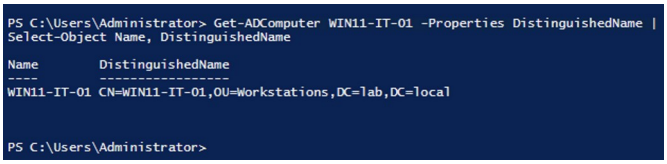
E01 - DC01 hostname, administrative context and server build output.



E02 - winver confirms Windows Server 2025 Standard Evaluation, version 24H2, build 26100.32230.



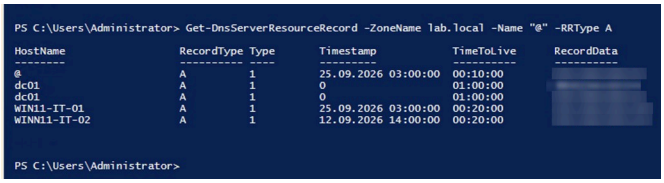
E03 - winver confirms Windows 11 Pro 25H2, build 26200.9457.



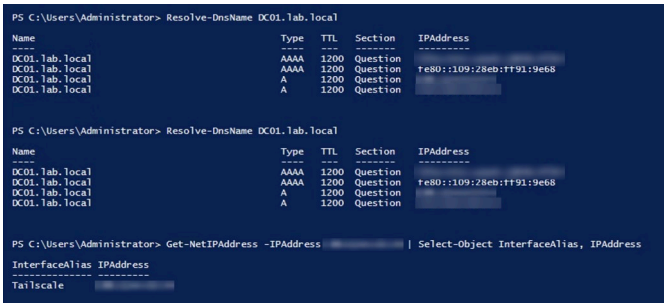
E04 - WIN11-IT-01 is in OU=Workstations,DC=lab,DC=local.

PASS	Server/client versions and the Workstations OU placement are shown in the evidence.

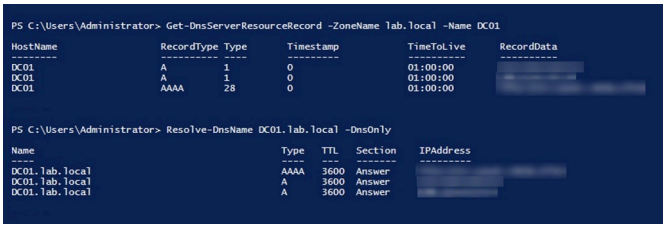
The main health checks passed. WIN11-IT-01 could find DC01 and the secure channel was healthy. One thing stood out: DC01 had both its lab IP and a Tailscale address (100.x) in DNS. When I checked, the PC had actually found DC01 through the Tailscale address. I manage the lab over Tailscale, so I left the records alone during P1 and wrote it down to clean up later.



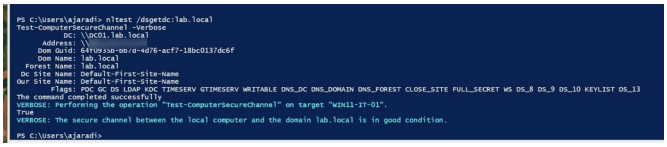
E05 - Current A records in lab.local, including DC01 and the client records.



E06 - Resolve-DnsName returns both the lab address and a Tailscale 100.x address; the adapter check confirms the 100.x address belongs to Tailscale.



E07 - The DC01 DNS record contains both the lab IP and a Tailscale 100.x address.



E08 - WIN11-IT-01 locates DC01.lab.local through the Tailscale 100.x address, and Test-ComputerSecureChannel reports a healthy relationship.

PASS	WIN11-IT-01 can find DC01 and the secure channel is healthy. The extra Tailscale DNS record is noted for later.

Task: Use a normal account for daily work and a separate account only when I actually need admin rights.

My normal desktop runs as LAB\ajaradi. The captured token is Medium integrity, and ajaradi is not in the local Administrators group. When I choose Run as administrator, Windows asks for another account. I use LAB\adm-ajaradi for that, and only the new PowerShell window runs elevated at High integrity.

```
PS C:\Users\ajaradi> whoami

whoami /groups

net localgroup Administrators
lab\ajaradi

GROUP INFORMATION
-----------------

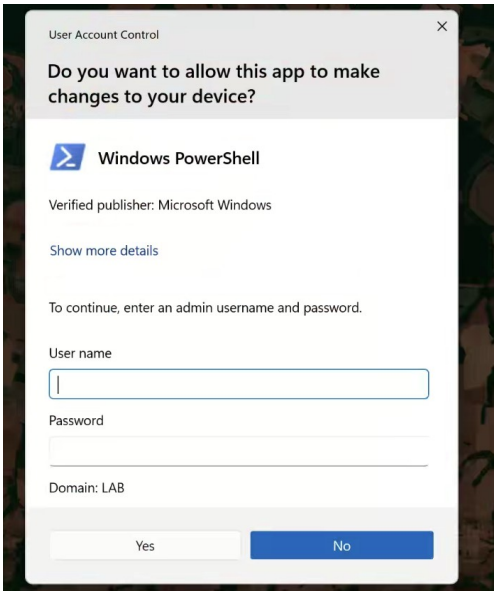
Group Name                                 Type             SID          Attributes
========================================== ================ ============ ==================================================
Everyone                                   Well-known group S-1-1-0      Mandatory group, Enabled by default, Enabled group
BUILTIN\Users                              Alias            S-1-5-32-545 Mandatory group, Enabled by default, Enabled group
NT AUTHORITY\INTERACTIVE                   Well-known group S-1-5-4      Mandatory group, Enabled by default, Enabled group
CONSOLE LOGON                              Well-known group S-1-2-1      Mandatory group, Enabled by default, Enabled group
NT AUTHORITY\Authenticated Users           Well-known group S-1-5-11     Mandatory group, Enabled by default, Enabled group
NT AUTHORITY\This Organization             Well-known group S-1-5-15     Mandatory group, Enabled by default, Enabled group
LOCAL                                      Well-known group S-1-2-0      Mandatory group, Enabled by default, Enabled group
Authentication authority asserted identity Well-known group S-1-18-1     Mandatory group, Enabled by default, Enabled group
Mandatory Label\Medium Mandatory Level     Label            S-1-16-8192
Alias name     Administrators
Comment        Administrators have complete and unrestricted access to the computer/domain

Members

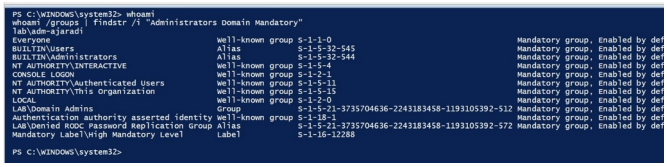
-------------------------------------------------------------------------------
Administrator
LAB\Domain Admins
LocalAdmin
The command completed successfully.


PS C:\Users\ajaradi>
```

E09 - Normal LAB\ajaradi shell: Medium integrity; ajaradi is not listed in the local Administrators group.



E10 - UAC asks the standard user for administrator credentials.

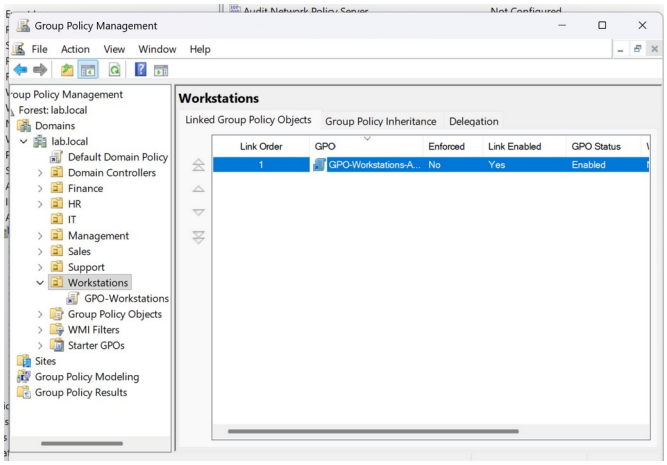


E11 - Elevated PowerShell runs as LAB\adm-ajaradi with High integrity and administrative group membership.

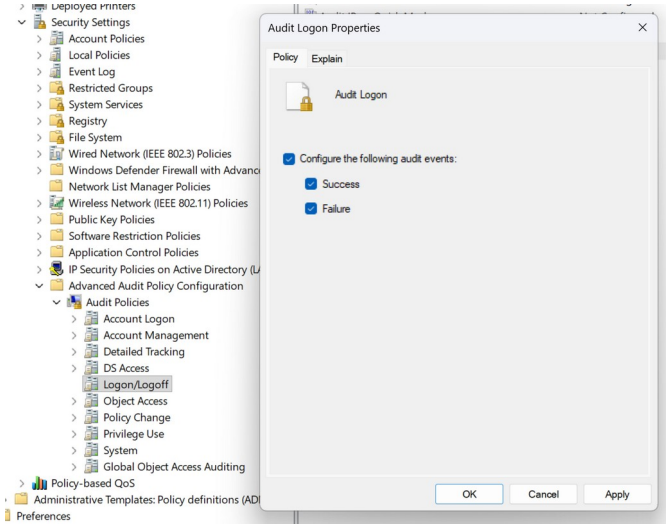
PASS	ajaradi stays a standard user. Admin work runs in a separate elevated process as adm-ajaradi.

Task: Record both successful and failed logons on WIN11-IT-01 and make sure the GPO actually reaches the client.

I created the audit GPO, linked it to the Workstations OU, refreshed policy on WIN11-IT-01, and checked auditpol. I wanted to see the Success and Failure setting on the PC itself, not only in the GPO console.



E12 - GPO-Workstations-Audit-Logon linked to the Workstations OU.



E13 - Audit Logon configured for Success and Failure.

```
PS C:\WINDOWS\system32> whoami
lab\adm-ajaradi
PS C:\WINDOWS\system32> gpupdate /target:computer /force
Updating policy...

Computer Policy update has completed successfully.

PS C:\WINDOWS\system32>
```

E14 - Admin context recorded; computer policy refresh completes successfully.

```
PS C:\WINDOWS\system32> auditpol /get /subcategory:"Logon"
System audit policy
Category/Subcategory                      Setting
Logon/Logoff
  Logon                                   Success and Failure
PS C:\WINDOWS\system32>
```

E15 - Effective local audit setting reports Logon = Success and Failure.

PASS	The GPO is configured and the client shows Logon auditing set to Success and Failure.

Next I used gpresult to make sure the GPO actually reached WIN11-IT-01. Then I used runas twice: first with a wrong password and then with the right one. That gave me one failed and one successful logon to look for in the Security log.

```
PS C:\WINDOWS\system32> gpresult /scope computer /r

Microsoft (R) Windows (R) Operating System Group Policy Result tool v2.0
© Microsoft Corporation. All rights reserved.

Created on  9/ 22/ 2026 at 12:26:41 AM


RSOP data for  on WIN11-IT-01 : Logging Mode
---------------------------------------------

OS Configuration:            Member Workstation
OS Version:                  10.0.26200
Site Name:                   Default-First-Site-Name
Roaming Profile:
Local Profile:
Connected over a slow link?: No


COMPUTER SETTINGS
------------------
    CN=WIN11-IT-01,OU=Workstations,DC=lab,DC=local
    Last time Group Policy was applied: 9/22/2026 at 12:22:59 AM
    Group Policy was applied from:      DC01.lab.local
    Group Policy slow link threshold:   500 kbps
    Domain Name:                        LAB
    Domain Type:                        Windows 2008 or later

    Applied Group Policy Objects
    -----------------------------
        GPO-Workstations-Audit-Logon
        Default Domain Policy

    The following GPOs were not applied because they were filtered out
    -------------------------------------------------------------------
        Local Group Policy
            Filtering:  Not Applied (Empty)

    The computer is a part of the following security groups
    -------------------------------------------------------
        BUILTIN\Administrators
        Everyone
        BUILTIN\Users
        NT AUTHORITY\NETWORK
        NT AUTHORITY\Authenticated Users
        This Organization
        WIN11-IT-01$
        Domain Computers
        Authentication authority asserted identity
        System Mandatory Level

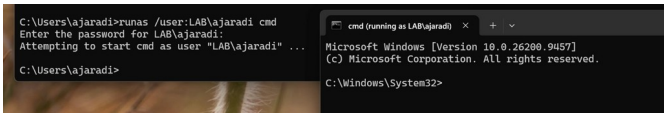
PS C:\WINDOWS\system32>
```

E16 - gpresult shows WIN11-IT-01 in the Workstations OU with the audit GPO applied.

```
C:\Users\ajaradi>runas /user:LAB\ajaradi cmd
Enter the password for LAB\ajaradi:
Attempting to start cmd as user "LAB\ajaradi" ...
RUNAS ERROR: Unable to run - cmd
1326: The user name or password is incorrect.

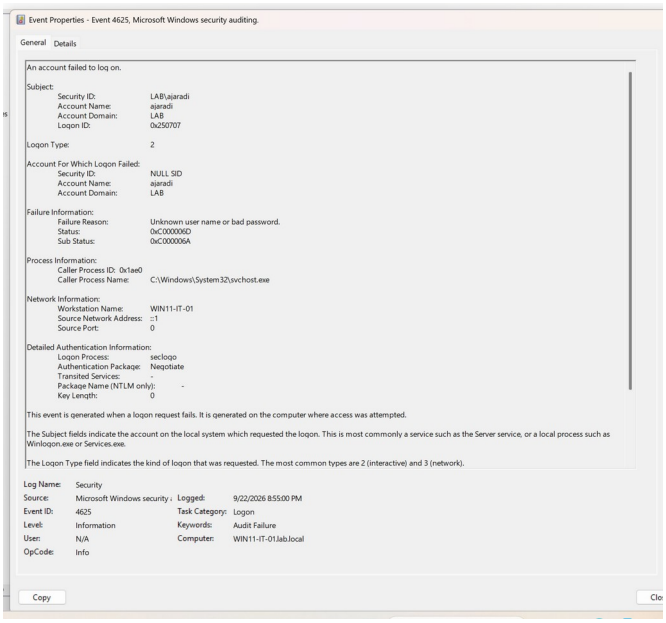
C:\Users\ajaradi>
```

E17 - Deliberately incorrect runas password; error 1326.

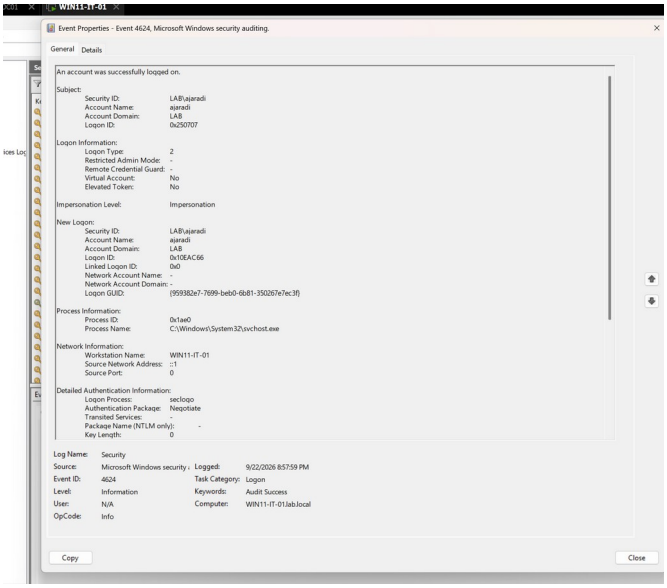


E18 - Correct credentials open a command window as LAB\ajaradi.

Event Viewer showed both results. The failed attempt is Event 4625, Logon Type 2, for LAB\ajaradi with the bad-password status values. The successful attempt is Event 4624, also Type 2, and the captured event shows Elevated Token = No.



E19 - Event 4625: failed Type 2 logon for LAB\ajaradi; status 0xC000006D, substatus 0xC000006A.



E20 - Event 4624: successful Type 2 logon for LAB\ajaradi; Elevated Token = No.

PASS	The policy is applied and the Security log contains both the failed and successful test logons.

Task: Let Jonas reset Finance passwords, but not HR passwords, and do it without making him an administrator.

I kept this inside PortfolioLab and used throwaway Finance and HR users. The permission went to the GG_PortfolioLab_Helpdesk group, not straight to Jonas. In the delegation wizard I gave that group only the password-reset task on the Finance OU.

```
PS C:\Users\Administrator>
New-ADOrganizationalUnit -Name "PortfolioLab" -Path "DC=lab,DC=local"

"Finance","HR","Disabled" | ForEach-Object {
    New-ADOrganizationalUnit -Name $_ -Path "OU=PortfolioLab,DC=lab,DC=local"
}

Get-ADOrganizationalUnit -Filter * -SearchBase "OU=PortfolioLab,DC=lab,DC=local" |
    Select-Object Name, DistinguishedName

Name          DistinguishedName
----          -----------------
PortfolioLab OU=PortfolioLab,DC=lab,DC=local
Finance       OU=Finance,OU=PortfolioLab,DC=lab,DC=local
HR            OU=HR,OU=PortfolioLab,DC=lab,DC=local
Disabled      OU=Disabled,OU=PortfolioLab,DC=lab,DC=local


PS C:\Users\Administrator>
```

E21 - PortfolioLab with Finance, HR and Disabled child OUs.

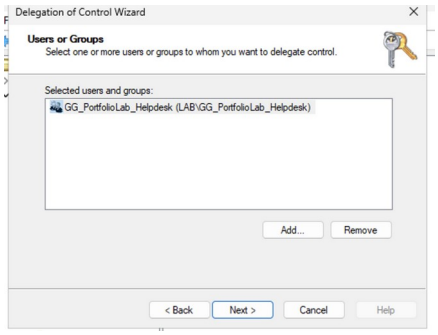
```
PS C:\WINDOWS\system32> Get-ADUser -Filter * `
-SearchBase "OU=PortfolioLab,DC=lab,DC=local" |
Select-Object Name, SamAccountName, DistinguishedName

Get-ADGroupMember "GG_PortfolioLab_Helpdesk" |
Select-Object Name, SamAccountName

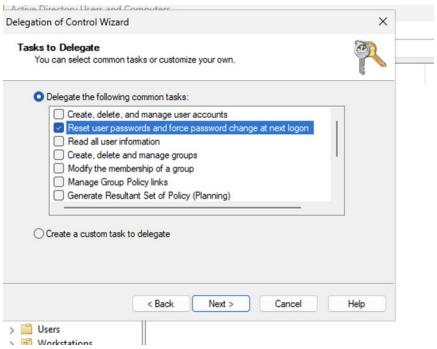
Name           SamAccountName DistinguishedName
----           -------------- -----------------
Finance Test01 finance.test01 CN=Finance Test01,OU=Finance,OU=PortfolioLab,DC=lab,DC=local
HR Test01      hr.test01      CN=HR Test01,OU=HR,OU=PortfolioLab,DC=lab,DC=local
Jonas Helpdesk jonas          CN=Jonas Helpdesk,OU=PortfolioLab,DC=lab,DC=local
Jonas Helpdesk jonas


PS C:\WINDOWS\system32> |
```

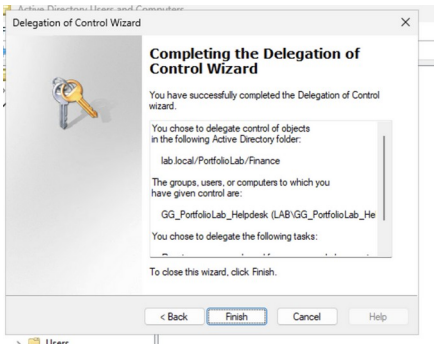
E22 - Finance Test01, HR Test01, Jonas Helpdesk and the helpdesk security group verified.



E23 - GG_PortfolioLab_Helpdesk selected in Delegation of Control.



E24 - Only Reset user passwords and force password change at next logon is selected.



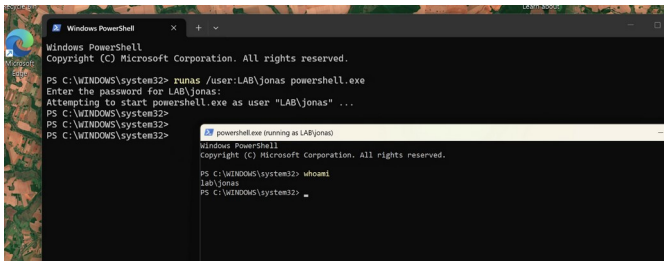
E25 - Wizard summary shows the Finance OU and the delegated helpdesk group.

I checked the Finance OU ACL, then tested the permission as LAB\jonas. I made one mistake first: I was still in an Administrator window, so that result proved nothing. I threw that test away and opened a PowerShell window as Jonas. From there the Finance reset worked, the HR reset was denied, and Jonas did not show Domain Admins or BUILTIN\Administrators in the captured group check.

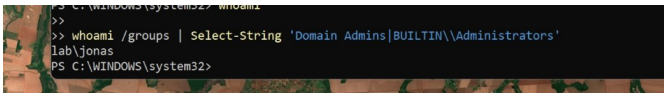
```
PS C:\WINDOWS\system32> dsacls "OU=Finance,OU=PortfolioLab,DC=lab,DC=local" | findstr /i "Helpdesk"
dsacls "OU=HR,OU=PortfolioLab,DC=lab,DC=local" | findstr /i "Helpdesk"
Allow LAB\GG_PortfolioLab_Helpdesk    Reset Password
Allow LAB\GG_PortfolioLab_Helpdesk    SPECIAL ACCESS for pwdLastSet

PS C:\WINDOWS\system32>
```

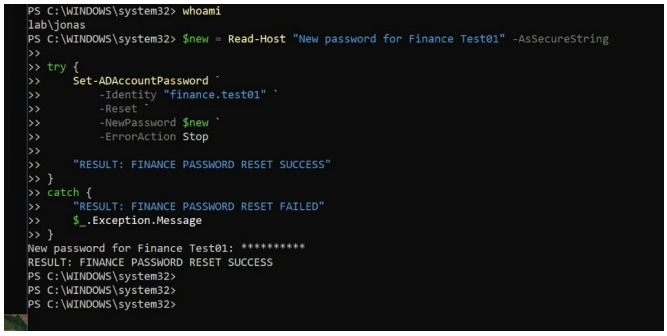
E26 - dsacls shows Reset Password and related pwdLastSet rights for the helpdesk group on the Finance OU.



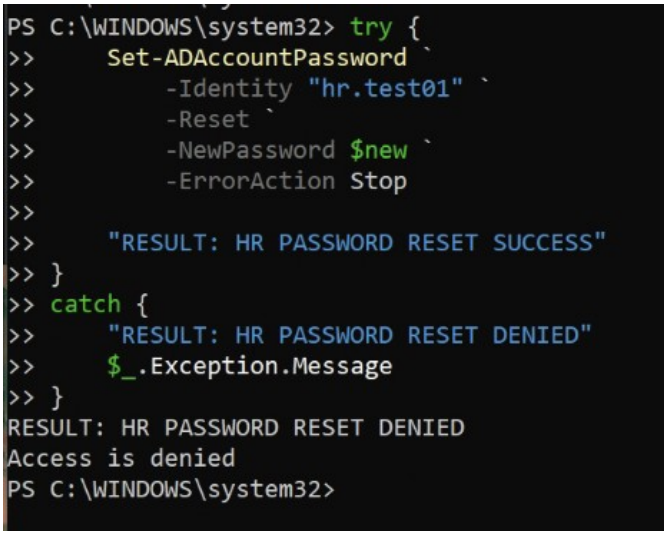
E27 - Test shell launched as LAB\jonas on WIN11-IT-01.



E28 - Captured Jonas token does not show Domain Admins or BUILTIN\Administrators.



E29 - As Jonas, Finance Test01 password reset succeeds.

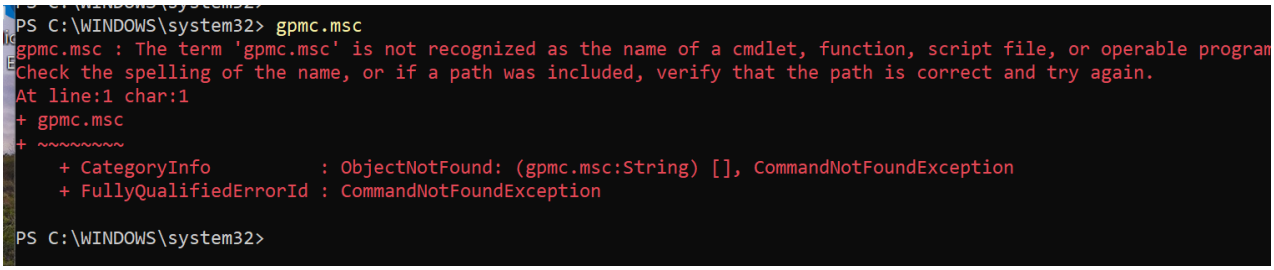


E30 - The same Jonas process receives Access is denied when resetting HR Test01.

PASS	Jonas can reset Finance Test01. The same reset against HR Test01 is denied.

Task: gpmc.msc does not open on WIN11-IT-01. Fix the admin tool and make sure it works afterward.

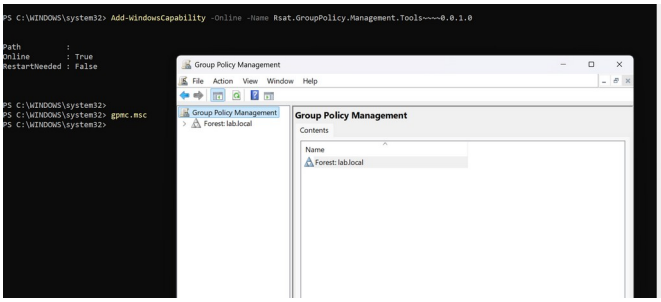
gpmc.msc was not recognised on WIN11-IT-01. I installed the RSAT Group Policy Management capability and tried again. GPMC opened normally after that. I didn't check first whether the tool was installed, so I can't prove it was missing, only that installing it fixed the problem.



E31 - Initial symptom: gpmc.msc is not recognised on WIN11-IT-01.



E32 - RSAT Group Policy Management capability installation completes; RestartNeeded = False.

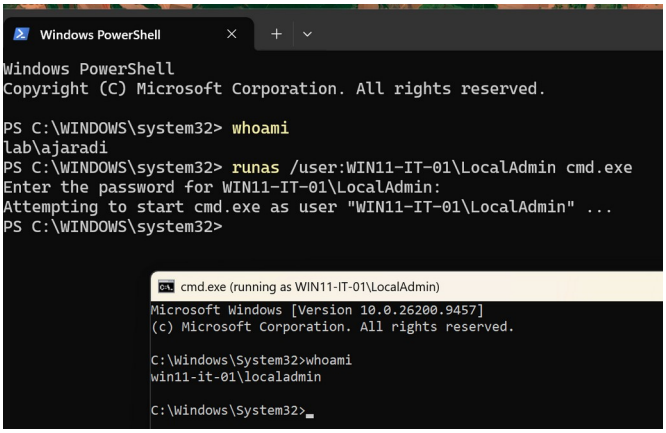


E33 - Retest: Group Policy Management opens and shows Forest: lab.local.

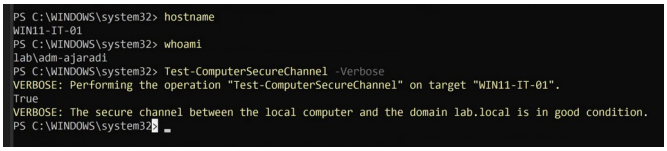
PASS	GPMC opens after the RSAT Group Policy Management capability is installed.

Task: Reproduce a broken workstation trust in the lab, keep LocalAdmin available as a fallback, then repair the secure channel without a domain rejoin.

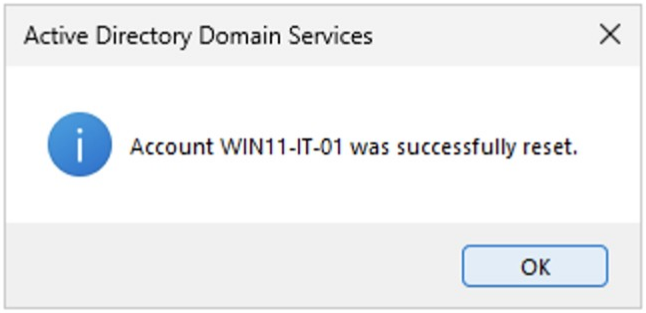
Before I broke anything, I checked the local LocalAdmin account and made sure it worked without the domain. I also captured the secure channel while it was still healthy. That gave me a way back into the client if the domain login stopped working.



E34 - Local recovery account verified before the test.



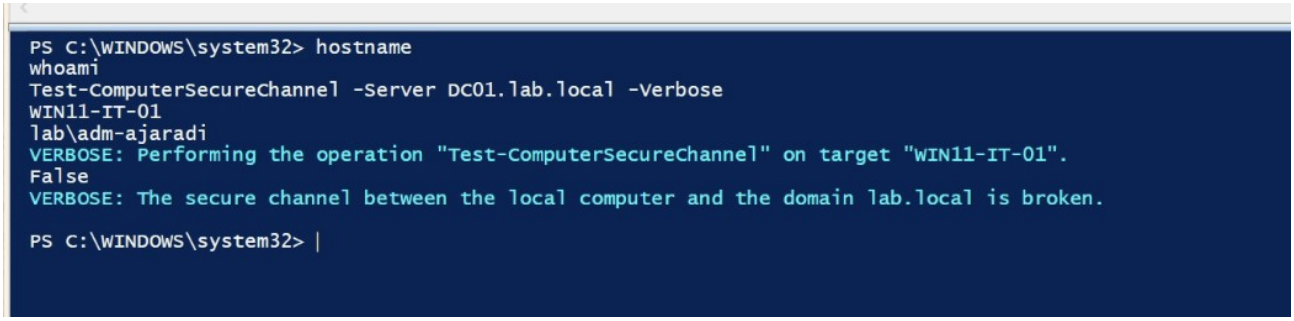
E35 - Baseline: WIN11-IT-01 secure channel returns True and reports good condition.



E36 - AD DS confirms that the WIN11-IT-01 computer account was reset.

This one took a while. Right after the computer-account reset, the open client session still reported a healthy channel. I did not count that as a failure. After I restarted WIN11-IT-01, the same test returned False and reported that the secure channel was broken.

After the restart, the client finally showed the trust problem: Test-ComputerSecureChannel returned False. I repaired it from the client with Test-ComputerSecureChannel -Repair and my separate admin credentials. Then I ran the test again instead of assuming the repair worked.

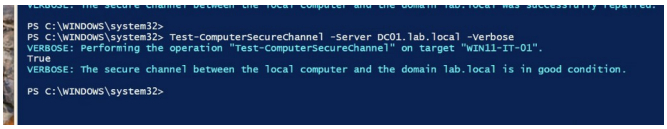


E37 - Broken state: Test-ComputerSecureChannel returns False and reports the channel is broken.

```
$cred = Get-Credential "LAB\adm-ajaradi"
Test-ComputerSecureChannel -Repair -Server DC01.lab.local -Credential $cred -Verbose
```



E38 - Repair command returns True and reports that the secure channel was successfully repaired.

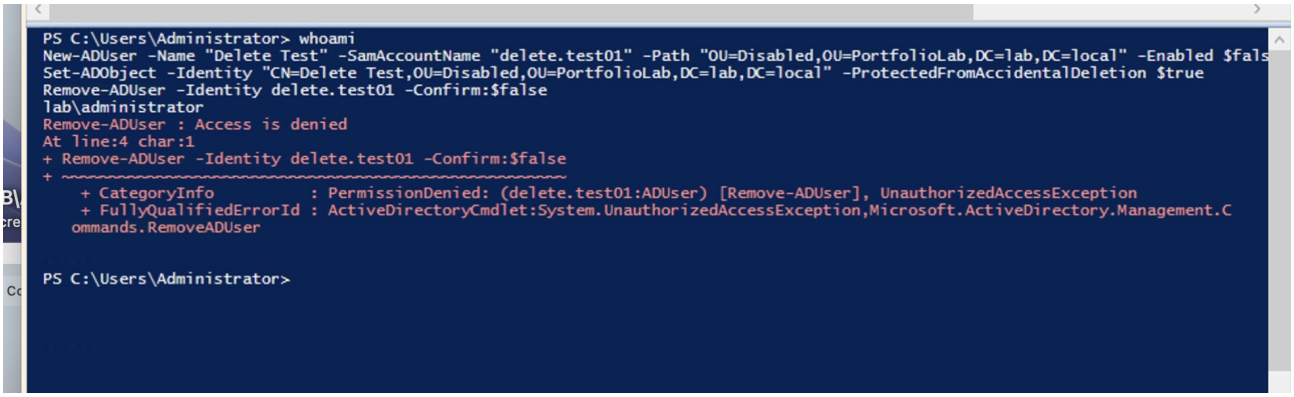


E39 - Independent retest returns True and reports the secure channel is in good condition.

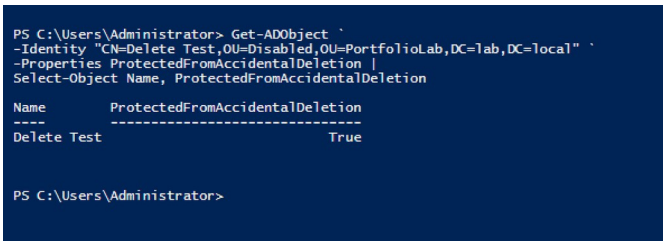
PASS	The secure channel is healthy again, and WIN11-IT-01 never had to leave lab.local.

Task: A disposable AD user returns Access is denied even from an admin session. Check the object before changing permissions.

The delete returned Access is denied even though I was using an admin account. Instead of changing permissions, I checked the object itself. ProtectedFromAccidentalDeletion was True. I turned that protection off, deleted the test user, and confirmed with Get-ADUser that the object was gone.



E40 - Initial delete attempt returns Access is denied.



E41 - ProtectedFromAccidentalDeletion is explicitly confirmed as True.



E42 - After removing protection and deleting the user, Get-ADUser reports the object cannot be found.

PASS	Accidental-deletion protection was on. After I turned it off, the delete worked and the user could no longer be found.

The part I found most useful was having to prove each change. I had to check whether the GPO really reached the PC, whether Jonas could do only the job I gave him, and whether the broken trust was actually repaired. The trust test took the longest because the failure did not show until I restarted the client.

What I practised

- Active Directory computer and user administration in lab.local
- DNS record inspection and domain-controller discovery
- Separate standard and administrative identities with UAC
- Group Policy configuration plus gpupdate, auditpol and gpresult verification
- Windows Security log review for Events 4624 and 4625
- Delegation of Control with a positive test and an out-of-scope negative test
- RSAT troubleshooting on a Windows 11 administrative client
- Secure-channel diagnosis and repair without a domain rejoin
- ProtectedFromAccidentalDeletion troubleshooting on a disposable AD object

Limitations

- This lab has one domain controller, so I did not test AD replication or redundancy.
- lab.local is only for this lab. In a real environment I would use a namespace based on a domain the organisation owns.
- Windows Server is an evaluation installation.
- DC01 still has both its lab IP and a Tailscale address (100.x) registered in DNS. I left that alone during P1 and wrote it down for later.
- For the GPMC case, I captured the error, the RSAT install and the working console, but not the feature state before I installed it.

Next

P2 will add a file server and move into Finance and HR share permissions, role changes and file recovery. I will keep the same domain so the projects build on each other.

Evidence note: All screenshots are from this lab. Passwords are not shown. E01-E42 are just labels so I can find the original screenshots later.